

课程编号: 100050220 北京理工大学 2022 一 2023 学年 第 一 学期

2021 级 复变函数与数理方程 课程试卷 A 卷

开课学院: 集成电路与电子学院

任课教师: _____

试卷用途: ☐ 期中 ☒ 期末 ☐ 补考 ☐ 重修

考试形式: ☐ 开卷 ☐ 半开卷 ☒ 闭卷

考试日期: 2022 年 12 月 5 日 所需时间: 120 分钟

考试允许带: _____ 入场

班级: _____ 学号: _____ 姓名: _____

考生承诺: “我确认本次考试是完全通过自己的努力完成的。”

考生签名: _____

题序	一	二	三	四	五	六	七	总分
满分	14	12	14	10	18	16	16	100
得分								
评卷人	张腾	董锡超	王彦华	李元昊	杨通	邓小英	邢冀川	

备用数据:

-----以下为试卷内容-----

一、填空题（每题 2 分，共 14 分）

1. 设 $z = \frac{1+i}{i-1}$ ，则 $z^{2022} = \underline{-1}$

2. 对数函数 $\text{Ln}(-4+4i)$ 的主值为 $\underline{\ln 4\sqrt{2} + i\frac{3\pi}{4}}$

3. 函数 $f(z) = \bar{z}$ 在复平面上的连续域为整个复平面，解析域为处处不解析

4. 设函数 $f(z) = \frac{z+e^z}{\cos z}$ 的泰勒展开式为 $\sum_{n=0}^{+\infty} c_n z^n$ ，其收敛半径 $R = \underline{\pi/2}$

5. $z=0$ 是 $\frac{z}{6\sin z + z(z^2-6)}$ 的 4 级极点。

6. 满足狄利克雷条件的周期函数的频谱是离散谱（离散谱/连续谱），满足狄利克雷和绝对可积条件的非周期函数的频谱是连续谱（离散谱/连续谱）。

7. 设函数 $f(t)$ 的傅氏变换为 $\mathcal{F}[f(t)] = F(\omega)$ ，则 $\mathcal{F}[8f(2t) + f(t-3)] =$

$4F\left(\frac{\omega}{2}\right) + e^{-3i\omega}F(\omega)$

二、简算题（每题 6 分，共 12 分）

1. 设 $z = x + iy$ ，若 $u(x, y) = x^2 - y^2 + 2x$ ，能否找到一个函数 $v(x, y)$ 使

$f(z) = u(x, y) + iv(x, y)$ 为解析函数？若能，求出使 $f(0) = 5i$ 的 $f(z)$ ；若不能，

说明理由。

解： $\frac{\partial u}{\partial x} = 2x + 2, \frac{\partial^2 u}{\partial x^2} = 2; \frac{\partial u}{\partial y} = -2y, \frac{\partial^2 u}{\partial y^2} = -2 \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

因此 $u(x, y)$ 是调和函数，能构成解析函数。

$$f'(z) = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} = 2x + 2 + 2iy = 2z + 2 \Rightarrow f(z) = z^2 + 2z + C$$

$$f(0) = 5i \Rightarrow C = 5i \Rightarrow f(z) = z^2 + 2z + 5i$$

2. 计算积分 $\int_C (z^2 + 3 - \pi e^{\pi \bar{z}}) dz$, 其中 C 是从原点沿直线到 $z_1 = 1$, 再从 z_1 沿直线到 $z_2 = 1+i$ 的线段。

解: 函数 $z^2 + 3$ 解析, 所以有 $\int_C (z^2 + 3) dz = \int_0^{1+i} (z^2 + 3) dz = \frac{z^3}{3} + 3z \Big|_0^{1+i} = \frac{7}{3} + \frac{11}{3}i$

$$\begin{aligned} \int_C -\pi e^{\pi \bar{z}} dz &= \int_0^1 -\pi e^{\pi x} dx + \int_0^1 -\pi e^{\pi(1-iy)} i dy \\ &= -e^{\pi x} \Big|_0^1 + e^{\pi} e^{-i\pi y} \Big|_0^1 = 1 - e^{\pi} - 2e^{\pi} = 1 - 3e^{\pi} \end{aligned}$$

$$\therefore \int_C (z^2 + 3 - \pi e^{\pi \bar{z}}) dz = \frac{7}{3} + \frac{11}{3}i + 1 - 3e^{\pi} = \frac{10}{3} - 3e^{\pi} + \frac{11}{3}i$$

三、计算积分 (每题 7 分, 共 14 分)

1. 求函数 $f(z) = \frac{ze^z}{(z-3)(z-1)^2}$ 在点 $z = \infty$ 的留数, 并计算 $\oint_C f(z) dz$, 其中 C 为正向

圆周 $|z| = 2$.

解: 扩充复平面有 3 个孤立奇点: 2 级极点 $z = 1$, 1 级极点 $z = 3$, 还有 $z = \infty$.

$$\text{Res}[f(z), 3] = \frac{ze^z}{(z-1)^2} \Big|_{z=3} = \frac{3}{4}e^3, \quad \text{Res}[f(z), 1] = \left(\frac{ze^z}{z-3} \right)' \Big|_{z=1} = -\frac{5}{4}e$$

$$\therefore \text{Res}[f(z), \infty] = -\text{Res}[f(z), 1] - \text{Res}[f(z), 2] = \frac{5}{4}e - \frac{3}{4}e^3$$

$$\oint_C f(z) dz = 2\pi i \cdot \text{Res}[f(z), 1] = -\frac{5}{2}\pi e i$$

2. 利用留数计算实积分 $\int_0^{+\infty} \frac{\cos x}{(x^2+1)(x^2+9)} dx$

解: 令 $f(z) = \frac{1}{(z^2+1)(z^2+9)} e^{iz}$, $f(z)$ 在上半平面有两个一级极点 $z = i$, $z = 3i$, 则

$$\begin{aligned} \int_0^{+\infty} \frac{\cos x}{(x^2+1)(x^2+9)} dx &= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\cos x}{(x^2+1)(x^2+9)} dx = \frac{1}{2} \text{Re} \{ 2\pi i \text{Res}[f(z), i] + 2\pi i \text{Res}[f(z), 3i] \} \\ &= \text{Re} \left(\pi i \frac{1}{(z+i)(z^2+9)} e^{iz} \Big|_{z=i} + \pi i \frac{1}{(z^2+1)(z+3i)} e^{iz} \Big|_{z=3i} \right) = \frac{\pi}{16} e^{-1} - \frac{\pi}{48} e^{-3} \end{aligned}$$

四、(10 分) 设函数 $f(z) = \frac{z-i}{(z-1)(z-2)}$

(1) 将 $f(z)$ 在以 $z=i$ 为中心的圆域内展开为泰勒级数, 并求其收敛域;

(2) 除泰勒展开式之外, $f(z)$ 在以 $z=i$ 为中心的圆环内洛朗展开式有几个? 试写

出各圆环域范围及各范围内的洛朗展开式.

解: (1) 离 $z=i$ 最近的奇点是 $z=1$, 因此泰勒级数的收敛半径 $R=|i-1|=\sqrt{2}$, 收敛域为 $|z-i|<\sqrt{2}$

$$\begin{aligned} f(z) &= \frac{z-i}{(z-1)(z-2)} = (z-i) \left(\frac{1}{z-2} - \frac{1}{z-1} \right) = (z-i) \left(\frac{1}{z-i-2+i} - \frac{1}{z-i-1+i} \right) \\ &= (z-i) \left(\frac{1}{i-2} \frac{1}{1+\frac{z-i}{i-2}} - \frac{1}{i-1} \frac{1}{1+\frac{z-i}{i-1}} \right) = (z-i) \left[\frac{1}{i-2} \sum_{n=0}^{+\infty} \left(-\frac{z-i}{i-2} \right)^n - \frac{1}{i-1} \sum_{n=0}^{+\infty} \left(-\frac{z-i}{i-1} \right)^n \right] = \\ &= \sum_{n=0}^{+\infty} (-1)^n \left[\left(\frac{1}{i-2} \right)^{n+1} - \left(\frac{1}{i-1} \right)^{n+1} \right] (z-i)^{n+1}, \quad |z-i|<\sqrt{2} \end{aligned}$$

(2) 洛朗展开式有 2 个

在 $\sqrt{2}<|z-i|<\sqrt{5}$ 内

$$\begin{aligned} f(z) &= (z-i) \left(\frac{1}{z-i-2+i} - \frac{1}{z-i-1+i} \right) = (z-i) \left(\frac{1}{i-2} \frac{1}{1+\frac{z-i}{i-2}} - \frac{1}{z-i} \frac{1}{1+\frac{i-1}{z-i}} \right) \\ &= (z-i) \left[\frac{1}{i-2} \sum_{n=0}^{+\infty} \left(-\frac{z-i}{i-2} \right)^n - \frac{1}{z-i} \sum_{n=0}^{+\infty} \left(-\frac{i-1}{z-i} \right)^n \right] = \sum_{n=0}^{+\infty} (-1)^n \left(\frac{1}{i-2} \right)^{n+1} (z-i)^{n+1} - \sum_{n=0}^{+\infty} (1-i)^n (z-i)^{-n} \end{aligned}$$

在 $\sqrt{5}<|z-i|<+\infty$ 内

$$f(z) = (z-i) \left(\frac{1}{z-i} \frac{1}{1+\frac{i-2}{z-i}} - \frac{1}{z-i} \frac{1}{1+\frac{i-1}{z-i}} \right) = \sum_{n=0}^{+\infty} \left[(2-i)^n - (1-i)^n \right] (z-i)^{-n}$$

五、求下列定解问题(共 2 题, 共 18 分)

$$1. \quad (9 \text{ 分}) \quad \begin{cases} \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial x \partial y} - 12 \frac{\partial^2 u}{\partial y^2} = 0, & -\infty < x < +\infty, y > 0 \\ \frac{\partial u}{\partial y} \Big|_{y=0} = 4e^x, \quad u \Big|_{y=0} = 0, & -\infty < x < +\infty \end{cases}$$

解：将方程标准化，其特征方程为

$$\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} - 12 = 0 \Rightarrow \begin{cases} \frac{dy}{dx} = -4 \Rightarrow 4x + y + c_1 = 0 \\ \frac{dy}{dx} = 3 \Rightarrow 3x - y + c_2 = 0 \end{cases}$$

做特征变换： $\xi = 4x + y$, $\eta = 3x - y$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial x} = 4 \frac{\partial u}{\partial \xi} + 3 \frac{\partial u}{\partial \eta},$$

$$\frac{\partial^2 u}{\partial x^2} = 4 \frac{\partial^2 u}{\partial \xi^2} \frac{\partial \xi}{\partial x} + 4 \frac{\partial^2 u}{\partial \xi \partial \eta} \frac{\partial \eta}{\partial x} + 3 \frac{\partial^2 u}{\partial \xi \partial \eta} \frac{\partial \xi}{\partial x} + 3 \frac{\partial^2 u}{\partial \eta^2} \frac{\partial \eta}{\partial x} = 16 \frac{\partial^2 u}{\partial \xi^2} + 24 \frac{\partial^2 u}{\partial \xi \partial \eta} + 9 \frac{\partial^2 u}{\partial \eta^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = 4 \frac{\partial^2 u}{\partial \xi^2} \frac{\partial \xi}{\partial y} + 4 \frac{\partial^2 u}{\partial \xi \partial \eta} \frac{\partial \eta}{\partial y} + 3 \frac{\partial^2 u}{\partial \xi \partial \eta} \frac{\partial \xi}{\partial y} + 3 \frac{\partial^2 u}{\partial \eta^2} \frac{\partial \eta}{\partial y} = 4 \frac{\partial^2 u}{\partial \xi^2} - \frac{\partial^2 u}{\partial \xi \partial \eta} - 3 \frac{\partial^2 u}{\partial \eta^2}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial y} = \frac{\partial u}{\partial \xi} - \frac{\partial u}{\partial \eta},$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial \xi^2} \frac{\partial \xi}{\partial y} + \frac{\partial^2 u}{\partial \xi \partial \eta} \frac{\partial \eta}{\partial y} - \frac{\partial^2 u}{\partial \xi \partial \eta} \frac{\partial \xi}{\partial y} - \frac{\partial^2 u}{\partial \eta^2} \frac{\partial \eta}{\partial y} = \frac{\partial^2 u}{\partial \xi^2} - 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}$$

代入原方程化简，得到： $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$

原方程的通解为： $u(x, y) = f_1(4x + y) + f_2(3x - y)$

代入边界条件： $\begin{cases} u \Big|_{y=0} = 0 = f_1(4x) + f_2(3x) \\ \frac{\partial u}{\partial y} \Big|_{y=0} = 4e^x = f_1'(4x) - f_2'(3x) \Rightarrow \frac{1}{4} f_1(4x) - \frac{1}{3} f_2(3x) = 4e^x + c \end{cases}$

解得： $\begin{cases} f_1(4x) = \frac{48}{7} e^x + \frac{12}{7} c \Rightarrow f_1(x) = \frac{48}{7} e^{\frac{x}{4}} + \frac{12}{7} c \\ f_2(3x) = -\frac{48}{7} e^x - \frac{12}{7} c \Rightarrow f_2(x) = -\frac{48}{7} e^{\frac{x}{3}} - \frac{12}{7} c \end{cases}$

$$\begin{aligned}
 u(x, y) &= f_1(4x+y) + f_2(3x-y) = \frac{48}{7} e^{\frac{4x+y}{4}} - \frac{48}{7} e^{\frac{3x-y}{3}} \\
 &= \frac{48}{7} e^x \left(e^{\frac{y}{4}} - e^{-\frac{y}{3}} \right)
 \end{aligned}$$

2. (9 分) 用积分变换法求解以下问题

$$\begin{cases} 2 \frac{\partial u(x, t)}{\partial t} = \frac{\partial^2 u(x, t)}{\partial x^2}, & -\infty < x < +\infty, t > 0 \\ u(x, 0) = \delta(x) - 2\delta(x+100) \end{cases}$$

注意：已知傅里叶变换对 $\delta(x) \leftrightarrow 1$, $\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} \leftrightarrow e^{-\frac{\omega^2\sigma^2}{2}}$

解：方程和初值条件做关于 x 的傅里叶变换，不妨设

$$u(x, t) \rightarrow U(\omega, t) = \int_{-\infty}^{+\infty} u(x, t) e^{-i\omega x} dx$$

$$u(x, 0) = \varphi(x)$$

原方程可以变为

$$\begin{cases} \frac{dU(\omega, t)}{dt} = \frac{1}{2} (i\omega)^2 U(\omega, t) = -\frac{1}{2} \omega^2 U(\omega, t) \\ U(\omega, 0) = \Phi(\omega) \end{cases}$$

$$U(\omega, t) = C e^{-\frac{\omega^2}{2}t}, \text{ 代入初值条件, 得到 } C = \Phi(\omega)$$

$$U(\omega, t) = C e^{-\frac{\omega^2}{2}t} = \Phi(\omega) e^{-\frac{\omega^2}{2}t}$$

$$u(x, t) = \mathcal{F}^{-1}(U(\omega, t)) = \mathcal{F}^{-1}(\Phi(\omega) e^{-\frac{\omega^2}{2}t}) = \varphi(x) * \mathcal{F}^{-1}(e^{-\frac{\omega^2}{2}t})$$

根据已知傅里叶变换对 $\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} \leftrightarrow e^{-\frac{\omega^2\sigma^2}{2}}$, 得到 $\mathcal{F}^{-1}(e^{-\frac{\omega^2}{2}t}) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$

$$u(x, t) = \varphi(x) * \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} = \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} \varphi(x-\tau) e^{-\frac{\tau^2}{2t}} d\tau$$

$$= \frac{1}{\sqrt{2\pi t}} \int_{-\infty}^{+\infty} [\delta(x-\tau) - 2\delta(x-\tau+100)] e^{-\frac{\tau^2}{2t}} d\tau = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} - \frac{2}{\sqrt{2\pi t}} e^{-\frac{(x+100)^2}{2t}}$$

另解: $U(\omega, t) = Ce^{-\frac{\omega^2}{2}t} = (1 - 2e^{j100\omega})e^{-\frac{\omega^2}{2}t}$, 根据傅里叶变换对以及时移性质

$$u(x, t) = \mathcal{F}^{-1} \left(e^{-\frac{\omega^2}{2}t} - 2e^{j100\omega} e^{-\frac{\omega^2}{2}t} \right) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} - \frac{2}{\sqrt{2\pi t}} e^{-\frac{(x+100)^2}{2t}}$$

六、分析计算题（共 2 题，共 16 分）

1. (8 分) 小强到一位研究贝塞尔函数的数学家的家中做客，想通过手机连接无线网络时，发现墙上贴着如下信息：

无线网络名称：BESSELFUNCTION，8 位数字密码为：*abcabcab*。已知

(1) $y(x) = J_0(1)Y_2(3)J_4(5)Y_3(x)$;

(2) $y(x) = \frac{b-x^2}{x^2} J_1(x) - \frac{4}{x} J_0(x)$; (3) $y(x) = \frac{1}{x^c} \frac{d}{dx} [x^c J_c(x)]$

都是满足方程 $x^2 y'' + xy' + (x^2 - a)y = 0$ 的一个解 (a, b, c 均为整数)

请根据贝塞尔方程的相关知识以及第一类贝塞尔函数的递推公式，帮助小强分析出网络的密码。（注： $J_m(x)$ 为 m 阶第一类贝塞尔函数， $Y_m(x)$ 为 m 阶第二类贝塞尔函数。）

解：解(1)本质上是 $Y_3(x)$ 的倍数，因此原方程应该是 3 阶贝塞尔方程， $a=9$

利用 $\mu J_\mu(x) = \frac{x}{2} [J_{\mu+1}(x) + J_{\mu-1}(x)]$ ，得到 $J_{\mu+1}(x) = 2\mu J_\mu(x) / x - J_{\mu-1}(x)$

$$J_3(x) = \frac{4}{x} J_2(x) - J_1(x)$$

$$J_2(x) = \frac{2}{x} J_1(x) - J_0(x)$$

$$\text{因此, } J_3(x) = \frac{8-x^2}{x^2} J_1(x) - \frac{4}{x} J_0(x), \quad b=8$$

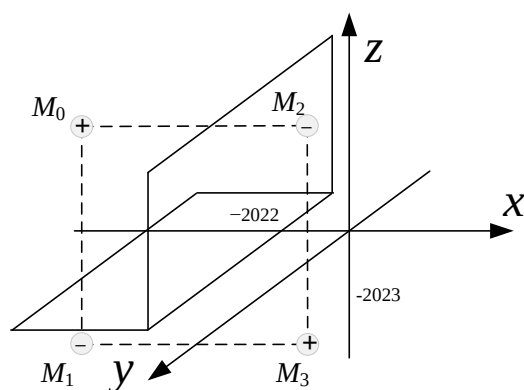
$$\text{根据 } \frac{d}{dx} [x^\mu J_\mu(x)] = x^\mu J_{\mu-1}(x), \quad c=4$$

密码：98498498

2. (8 分)用格林函数法求下列定解问题（结果只写出形式解即可）

$$\begin{cases} \Delta u = 0, & (x < -2022, -\infty < y < +\infty, z > -2023) \\ u|_{z=-2023} = g(x, y), & (x < -2022, -\infty < y < +\infty) \\ u|_{x=-2022} = h(y, z), & (z > -2023, -\infty < y < +\infty) \end{cases}$$

解：



如图在所求区域放置正电荷，则根据电像法有 3 个像点，其空间域格林函数为

$$G = \frac{1}{4\pi} \left(\frac{1}{r_{MM_0}} - \frac{1}{r_{MM_1}} - \frac{1}{r_{MM_2}} + \frac{1}{r_{MM_3}} \right)$$

其沿平行于 XOY 平面的边界外法向偏导为：

$$\left. \frac{\partial G}{\partial \vec{n}_0} \right|_S = - \left. \frac{\partial G}{\partial z_0} \right|_{z_0 = -2023}$$

其沿平行于 YOZ 平面的边界外法向偏导为：

$$\left. \frac{\partial G}{\partial \vec{n}_0} \right|_S = \left. \frac{\partial G}{\partial x_0} \right|_{x_0 = -2022}$$

$$\begin{aligned} \therefore u(M) &= - \iint_S f(M_0) \frac{\partial G}{\partial \vec{n}_0} dS_0 \\ &= - \int_{-2023}^{+\infty} \int_{-\infty}^{+\infty} h(y_0, z_0) \left. \frac{\partial G}{\partial x_0} \right|_{x_0 = -2022} dy_0 dz_0 + \int_{-\infty}^{+\infty} \int_{-\infty}^{-2022} g(x_0, y_0) \left. \frac{\partial G}{\partial z_0} \right|_{z_0 = -2023} dx_0 dy_0 \end{aligned}$$

七、分析计算题（共 2 题，共 16 分）

1. （4 分）对下列圆域拉普拉斯定解问题，设其可分离变量形式的解为

$u(r, \varphi) = R(r)\Phi(\varphi)$ ，试补足求解该定解问题所需的隐含条件，并写出分离出来的本征值问题。

$$\begin{cases} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \varphi^2} = 0 & (0 < r < r_0, 0 < \varphi < 2\pi) \\ u|_{r=r_0} = f(\varphi) & (0 \leq \varphi < 2\pi) \end{cases}$$

解：

自然周期条件： $\Phi(0) = \Phi(2\pi), \left. \frac{d\Phi}{d\varphi} \right|_{\varphi=0} = \left. \frac{d\Phi}{d\varphi} \right|_{\varphi=2\pi}$

有界性条件： $|R(0)| < \infty$

本征值问题：
$$\begin{cases} \Phi'' + \lambda\Phi = 0 \\ \Phi(0) = \Phi(2\pi) \\ \Phi'(0) = \Phi'(2\pi) \end{cases}$$

2. (12 分) 某长度为 1 的均匀杆， $x=0$ 端固定， $x=1$ 端自由，各质点初始位移为 0，

初速度为 $x(1-x)$ ，其纵振动满足偏微分方程 $\frac{\partial^2 u(x, t)}{\partial t^2} = \frac{\partial^2 u(x, t)}{\partial x^2} + 2$ ，试写出该定解问题，并用分离变量法求解该定解问题。

注意：设 $\int_0^1 x \sin\left(n + \frac{1}{2}\right) \pi x dx = A_n$ ， $\int_0^1 x^2 \sin\left(n + \frac{1}{2}\right) \pi x dx = B_n$ ，无需计算这两个积分，

分别用 A_n, B_n 代替。

解：根据题意可得定解问题为

$$\frac{\partial^2 u(x, t)}{\partial t^2} = \frac{\partial^2 u(x, t)}{\partial x^2} + 2, \quad 0 < x < 1, t > 0$$

$$\text{初始条件: } u(x, 0) = 0, \quad \frac{\partial u(x, 0)}{\partial t} = x(1-x), \quad 0 \leq x \leq 1$$

$$\text{边界条件: } u(0, t) = 0, \quad \frac{\partial u(1, t)}{\partial x} = 0, \quad t > 0$$

本征函数展开法：设解为 $u(x, t) = X(x)T(t)$ ，第 I, II 类齐次边界条件，则本征值和本征函数为：

$$\lambda_n = \left[\left(n + \frac{1}{2} \right) \pi \right]^2, \quad X_n(x) = \sin \left[\left(n + \frac{1}{2} \right) \pi x \right], \quad n = 0, 1, 2, \dots$$

将解按本征函数展开 $u(x, t) = \sum_{n=0}^{\infty} T_n(t) \sin\left(n + \frac{1}{2}\right) \pi x$,

初始条件: $u(x, 0) = \sum_{n=1}^{\infty} T_n(0) \sin\left(n + \frac{1}{2}\right) \pi x = 0 \Rightarrow T_n(0) = 0$

$$\frac{\partial u(x, 0)}{\partial t} = \sum_{n=1}^{\infty} T'_n(0) \sin\left(n + \frac{1}{2}\right) \pi x = x(1-x)$$

$$T'_n(0) = 2 \int_0^1 x(1-x) \sin\left(n + \frac{1}{2}\right) \pi x dx = 2(A_n - B_n)$$

把非齐次项 2 可按照本征函数展开, 即 $2 = \sum_{n=1}^{\infty} f_n(t) \sin\left(n + \frac{1}{2}\right) \pi x$, 则

$$f_n(t) = 2 \int_0^1 2 \sin\left(n + \frac{1}{2}\right) \pi \xi d\xi = \frac{4}{\left(n + \frac{1}{2}\right) \pi}$$

$$\sum_{n=1}^{\infty} T''_n(t) \sin\left(n + \frac{1}{2}\right) \pi x = \sum_{n=1}^{\infty} \left[-\left(n + \frac{1}{2}\right)^2 \pi^2 T_n(t) \sin\left(n + \frac{1}{2}\right) \pi x \right] + \sum_{n=1}^{\infty} \frac{8}{(2n+1)\pi} \sin\left(n + \frac{1}{2}\right) \pi x$$

$$T''_n(t) + \left(n + \frac{1}{2}\right)^2 \pi^2 T_n(t) = \frac{4}{\left(n + \frac{1}{2}\right) \pi}, T_n(0) = 0, T'_n(0) = 2(A_n - B_n)$$

$$T_n(t) = C_1 \cos\left(n + \frac{1}{2}\right) \pi t + C_2 \sin\left(n + \frac{1}{2}\right) \pi t + \frac{4}{\left(n + \frac{1}{2}\right)^3 \pi^3}$$

$$\Rightarrow C_1 = \frac{-4}{\left(n + \frac{1}{2}\right)^3 \pi^3}, C_2 = 2(A_n - B_n) \left(n + \frac{1}{2}\right) \pi$$

$$T_n(t) = \frac{-4}{\left(n + \frac{1}{2}\right)^3 \pi^3} \cos\left(n + \frac{1}{2}\right) \pi t + 2(A_n - B_n) \left(n + \frac{1}{2}\right) \pi \sin\left(n + \frac{1}{2}\right) \pi t + \frac{4}{\left(n + \frac{1}{2}\right)^3 \pi^3}$$

$$\therefore u(x, t) = \sum_{n=0}^{\infty} T_n(t) \sin\left(n + \frac{1}{2}\right) \pi x$$

辅助函数法:

$$u(x, t) = v(x, t) + w(x), \text{ 则 } v_{tt} = [v_{xx} + w''(x)] + 2$$

$$\begin{cases} w''(x) + 2 = 0 \\ w|_{x=0} = 0, w'|_{x=1} = 0 \end{cases}$$

$$\therefore w(x) = -x^2 + 2x \Rightarrow u(x, t) = v(x, t) - x^2 + 2x$$

$$\begin{cases} \frac{\partial^2 v}{\partial t^2} = \frac{\partial^2 v}{\partial x^2}, 0 < x < 1, t > 0 \\ v|_{x=0} = 0, v_x|_{x=1} = 0 \\ v|_{t=0} = x^2 - 2x, \frac{\partial v}{\partial t}|_{t=0} = x(1-x) \end{cases}$$

$$v(x, t) = \sum_{n=0}^{\infty} \left[C_n \cos\left(n + \frac{1}{2}\right)\pi t + D_n \sin\left(n + \frac{1}{2}\right)\pi t \right] \sin\left(n + \frac{1}{2}\right)\pi x$$

$$C_n = 2 \int_0^1 (x^2 - 2x) \sin\left(n + \frac{1}{2}\right)\pi x dx = 2B_n - 4A_n$$

$$D_n = \frac{2}{n\pi} \int_0^1 x(1-x) \sin\left(n + \frac{1}{2}\right)\pi x dx = \frac{2}{n\pi} (A_n - B_n)$$

$$\therefore u(x, t) = v(x, t) - x^2 + 2x$$

$$= \sum_{n=0}^{\infty} \left[(2B_n - 4A_n) \cos\left(n + \frac{1}{2}\right)\pi t + \frac{2}{n\pi} (A_n - B_n) \sin\left(n + \frac{1}{2}\right)\pi t \right] \sin\left(n + \frac{1}{2}\right)\pi x - x^2 + 2x$$

本征函数展开法

$$T_n(t) = \frac{-4}{\left(n + \frac{1}{2}\right)^3 \pi^3} \cos\left(n + \frac{1}{2}\right)\pi t + 2(A_n - B_n) \left(n + \frac{1}{2}\right)\pi \sin\left(n + \frac{1}{2}\right)\pi t + \frac{4}{\left(n + \frac{1}{2}\right)^3 \pi^3}$$

$$\therefore u(x, t) = \sum_{n=0}^{\infty} T_n(t) \sin\left(n + \frac{1}{2}\right)\pi x$$